Situation & Task (The Problem)

As Supervisor, Onsite IT & Automation Systems, I identified a gap in how we handled password and account expiration for ~2,600 users in on-premises Active Directory, synced to Entra ID through Entra Connect. Expirations were handled reactively: people usually found out when they could no longer sign in. Some of this was done by hand, but only on a very limited scale. Doing it for all ~2,600 users (checking expiration dates, contacting users, following up with managers) **would have taken an estimated 380 hours a year**, so it **couldn't be sustained at that scale.**

Action (The Solution)

I scoped and built a set of **scheduled PowerShell scripts** against on-premises Active Directory that warn users, and their managers where needed, before a password or account expires.

- **Automated Detection:** Scheduled runs query AD for passwords and accounts approaching expiration within defined warning windows.
- **Proactive Communication:** Built a manager-aware notification framework that sent **9,000+ proactive notifications**, so users, and managers where their action was needed, could act before access was lost.
- **Policy Alignment:** Warning windows and rules follow the organization's Security policies for password and account expiration.
- **Audit-Ready Logging:** Every run and every notification is logged, so it is always possible to show who was warned and when.

Result (The Impact)

The automation replaced reactive lockouts with proactive warnings:

- **Coverage Extended:** from a limited manual process to all ~2,600 users.
- **Effort Covered:** an estimated **380 hours a year** of manual tracking and follow-up handled by automation.
- **Audit Trail:** every warning is logged, so it is always possible to show who was warned and when.